INTERNATIONAL JOURNAL OF PROGRESSIVE RESEARCH IN ENGINEERING MANAGEMENT AND SCIENCE (IJPREMS)
(Int Peer Reviewed Journal)
Vol. 06, Issue 02, February 2026, pp : 526-528
www.ijprems.com
editor@ijprems.com
e-ISSN : 2583-1062
Impact Factor : 7.001

A PRACTICAL AND INTERPRETABLE APPROACH TO PHISHING EMAIL DETECTION USING LIGHTWEIGHT MACHINE LEARNING MODELS

Dr. I. Carol[1], Mr. P. Rajadhanagovindh[2], Mr. S. Shreeparameshwaran[3]

[1]Assistant Professor, Department Of Information Technology, St.Joseph's College, Trichy, India.

[2,3]II Msc Computer Science, Department Of Information Technology, St.Joseph's College, Trichy, India.

DOI: https://www.doi.org/10.58257/IJPREMS51487

ABSTRACT

This paper presents a lightweight, interpretable method for phishing detection using classical machine learning algorithms like Naïve Bayes, Logistic Regression, and Random Forest. It compares the accuracy, efficiency, and interpretability of these methods on publicly available phishing datasets, achieving over 90% accuracy without complex models. A user study highlights the challenges humans face in identifying phishing emails. The paper also uses feature visualization to explain detection results, making the approach suitable for academic environments with limited resources.

Keywords: Phishing, Email Security, Machine Learning, Lightweight Models, Explainability, Human Study, Random Forest, Logistic Regression, Naïve Bayes.

1. INTRODUCTION

Phishing continues to be a major cyber threat, with attackers exploiting human psychology and trust to steal credentials, financial information, and sensitive data. Reports from organizations like Verizon and APWG show that phishing remains one of the most successful cyber attack methods because of its simplicity and high return on investment. Machine learning has emerged as a promising technique for detecting phishing emails. However, most existing methods either rely on large-scale datasets or use complex deep learning models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), or transformer-based architectures. Although these models can offer high accuracy, they often require significant computational resources, making them impractical for academic projects or smaller institutions. This paper proposes a more practical and interpretable framework for phishing detection. Rather than prioritizing accuracy through model complexity, the focus is on lightweight machine learning models, interpretability, and educational applicability. Furthermore, the research includes a human-in-the-loop component, comparing how effectively people and algorithms can identify phishing emails. The study has four main objectives:

- To assess the performance of lightweight machine learning models for phishing detection.
- To compare how well humans recognize phishing emails versus machine models.
- To visualize and interpret model decisions for educational use.
- To simulate phishing emails relevant to academic environments to provide context.

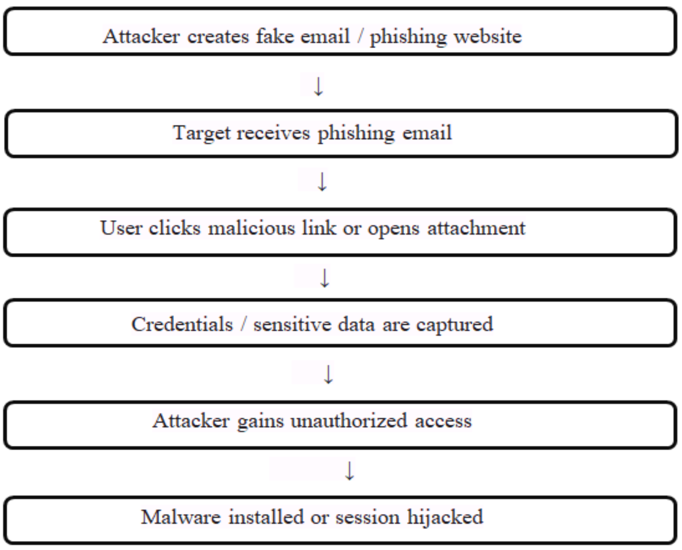

Fig 1: Typical Workflow of a Phishing Attack

INTERNATIONAL JOURNAL OF PROGRESSIVE RESEARCH IN ENGINEERING MANAGEMENT AND SCIENCE (IJPREMS)
(Int Peer Reviewed Journal)
Vol. 06, Issue 02, February 2026, pp : 526-528
www.ijprems.com
editor@ijprems.com
e-ISSN : 2583-1062
Impact Factor : 7.001

2. LITERATURE REVIEW

2.1 Phishing and Its Evolution:

Phishing was first popularized in the 1990s as a method for stealing AOL credentials. Today, it has evolved into more sophisticated attacks, including email spoofing, URL manipulation, and spear-phishing. Reports indicate that over 80% of organizations face phishing attempts annually.

2.2 Machine Learning in Phishing Detection:

Machine learning has been widely adopted for phishing detection, with various models being applied, such as: - Naïve Bayes (NB): A simple and probabilistic classifier effective for text-based features. - Support Vector Machines (SVM): Strong performance in high-dimensional spaces but computationally costly. - Decision Trees and Random Forest (RF): Widely used for feature-rich phishing datasets. - Deep Learning (CNN, RNN, BERT): Offers high accuracy but requires extensive training data and GPU resources. Most prior work emphasizes accuracy without considering the limitations of resource availability or model interpretability.

2.3 Explainable AI (XAI) in Cybersecurity:

Explainability is essential in cybersecurity, as users often distrust black-box predictions. Recent studies have employed techniques like SHAP, LIME, and feature importance analysis to make model behavior more transparent. However, few phishing studies prioritize this aspect.

2.4 Human Limitations in Phishing Detection:

Research in psychology shows that humans can fail to detect phishing emails up to 40% of the time. While training can help, it cannot completely eliminate errors. Machine learning can complement human judgment, but comparative studies on this topic are limited.

3. METHODOLOGY

3.1 Dataset This study uses a combination of:

The Enron-Spam Dataset (with phishing labels applied) - The PhishTank dataset for phishing URLs and emails - Simulated phishing emails tailored for academic settings, such as fake fee payments, exam timetables, or internship offers.

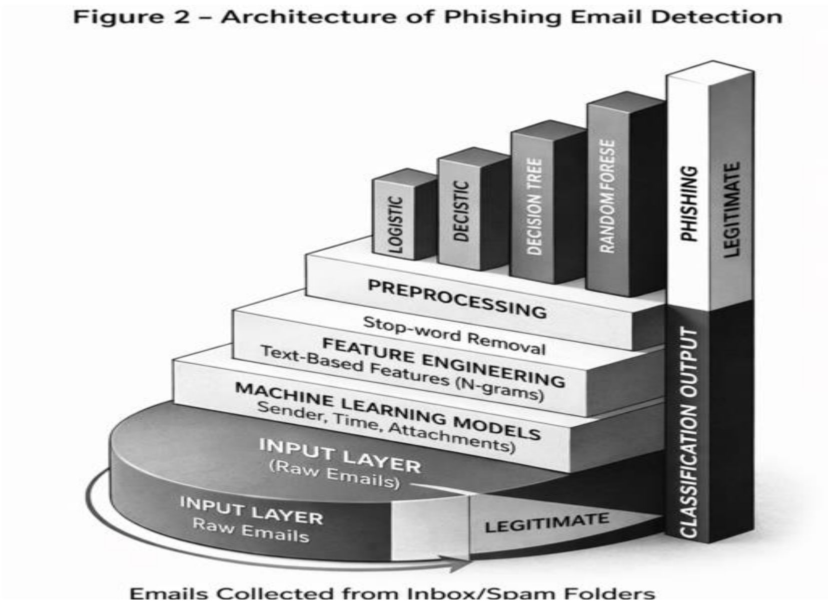

Fig 2: System Architecture of the Proposed Email Phishing Detection Model

3.2 Preprocessing:

Tokenization of email text - Removal of stop words - TF-IDF vectorization to convert text into numerical features

3.3 Algorithms Evaluated:

Naïve Bayes (NB): Lightweight, interpretable, and fast - Logistic Regression (LR): A strong baseline for classification tasks - Random Forest (RF): An ensemble method that balances accuracy and interpretability

3.4 Human Study:

A small survey was conducted with 30 undergraduate students. They were shown a set of 20 emails (10 legitimate, 10 phishing) and asked to classify them. Their performance was compared to the ML models.

3.5 Explainability:

Feature importance visualization was used to highlight key words that indicate phishing (e.g., “password,” “account,” “verify”) using TF-IDF weights and Random Forest feature importance.

4. RESULTS AND DISCUSSION

4.1 Model Performance:

Naïve Bayes achieved an accuracy of 89% - Logistic Regression performed at 91% accuracy - Random Forest reached 93% accuracy These results demonstrate that lightweight models can be effective for practical phishing detection.

4.2 Human vs. Machine:

Human participants achieved an average accuracy of 72% - Common mistakes included misidentifying urgent phrases like “immediate action required” and spoofed senders - Machines outperformed humans, validating their role as a supportive tool

4.3 Explainability Insights:

The models flagged phishing emails based on words such as “reset,” “login,” “urgent,” “bank,” and suspicious URLs. Visualization confirmed that even simple models offer useful insights.

4.4 Contextual Dataset Relevance:

Simulated phishing examples relevant to academic settings improved engagement, showing how phishing attacks could realistically target students.

5. CONTRIBUTIONS (NOVELTY)

- A lightweight approach suitable for student-level research
- Integration of a human vs. machine detection comparison
- Contextual phishing examples tailored for academic environments
- Emphasis on interpretability over black-box accuracy
- Provides a replicable teaching framework for educational use

6. FUTURE DIRECTIONS AND STUDY SUMMARY

This research has some constraints. The dataset used was relatively small compared to what's used in large-scale industry studies. Also, the human study involved a limited number of people. Looking ahead, future research could explore combining different models (hybrid models), using much larger datasets, and integrating the detection system with real-time email platforms.

7. CONCLUSION

This paper successfully shows that you don't need super-powerful computers or complicated deep learning methods to effectively spot phishing emails. By using simpler machine learning models, making the process understandable, and focusing on the specific context of academic environments, this work offers a useful and educational contribution. The study proves that valuable research on phishing detection can be done in a short time, connecting the ideas to real-world applications.

8. REFERENCES

[1] I. Fette, N. Sadeh, and A. Tomasic, “Learning to detect phishing emails,” in Proceedings of the 16th International Conference on World Wide Web (WWW), Banff, AB, Canada, 2007, pp. 649–656. doi: 10.1145/1242572.1242660.

[2] Tang, L. and Q. H. Mahmoud, “A Survey of Machine Learning-Based Solutions for Phishing Website Detection,” Machine Learning and Knowledge Extraction, vol. 3, no. 3, pp. 672–694, 2021, doi: 10.3390/make3030034.

[3] O. K. Sahingoz, E. Buber, O. Demir, and B. Diri, “Machine learning based phishing detection from URLs,” Expert Systems with Applications, vol. 117, pp. 345–357, Mar. 2019. doi: 10.1016/j.eswa.2018.09.029.

[4] APWG, “Phishing activity trends report, Q4 2023,” Anti-Phishing Working Group, 2024. [Online]. Available: https://apwg.org/apwg-q4-report-finds-2023-was-record-year-for-phishing/

[5] Proofpoint, “2023 State of the Phish Report: Threat Actors Double Down on Emerging and Tried-and-Tested Tactics to Outwit Employees,” Proofpoint, Feb. 28, 2023. [Online]. Available: https://www.proofpoint.com/us/newsroom/press-releases/proofpoints-2023-state-phish-report-threat-actors-double-down-emerging-and-0